Domain and range

1

a Any value of x.

b Only non-negative values of y.

2

a 0

b There is no greatest value.

c $y \geq 0$

d All real x

3

a 0

b There is no greatest value.

c $y \geq 0$

d All real x

4 Example answer: $f\left(x\right) = |x| + 2$

5

a $[0, 6]$

b $(3, 6]$

6

a

- i All real x
- ii $y \leq 0$

b

- i All real x
- ii $y \geq -1$

c

- i All real x
- ii $y \leq 4$

d

- i All real x
- ii $y \leq 2$

7

a

- i $[-5, -2]$
- ii $[0, 6]$

b

- i $[-2, 3)$
- ii $[-9, 0]$

c

- i $(-2, 1)$
- ii $[0, 6)$

d

- i $[2, 4)$
- ii $(0, 4]$

Graphs of absolute value functions

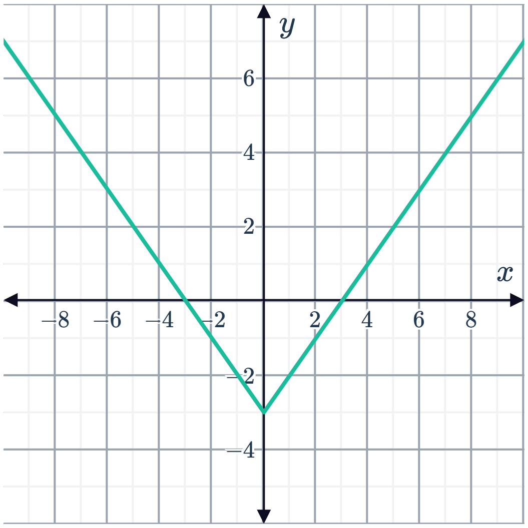
8

a

i All real x

ii $y \geq -3$

iii

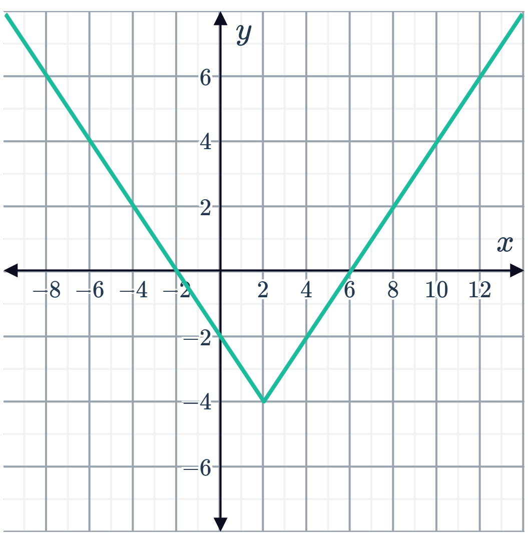


b

i All real x

ii $y \geq -4$

iii

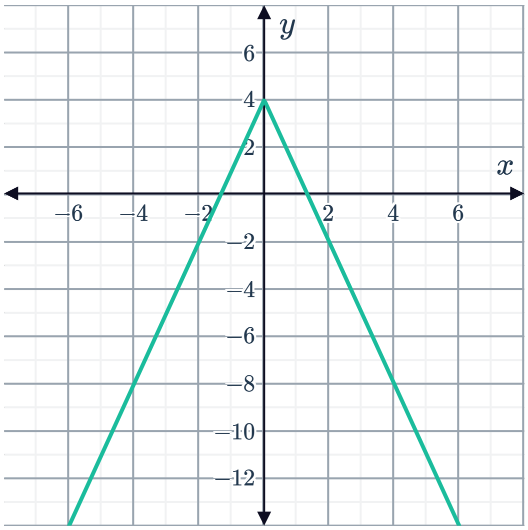


c

i All real x

ii $y \leq 4$

iii

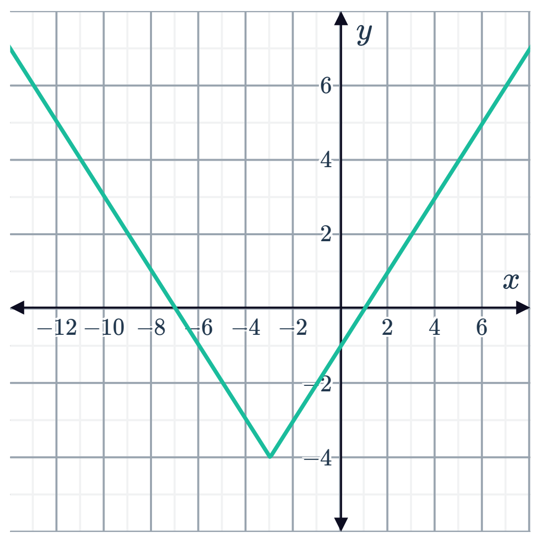


9

a

i $y \geq -4$

ii



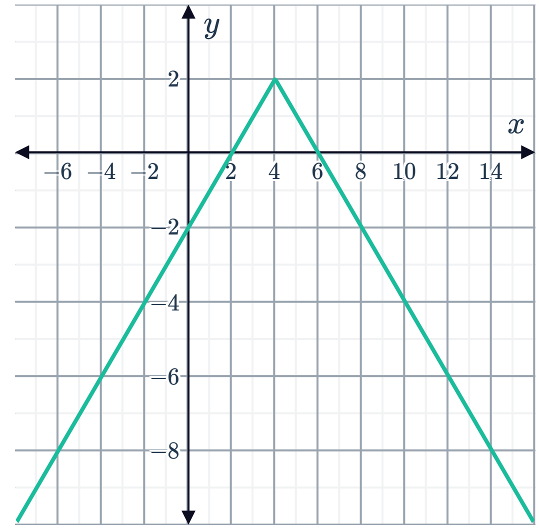
iii $(-3, -4)$

iv -4

b

i $y \leq 2$

ii



iii $(4, 2)$

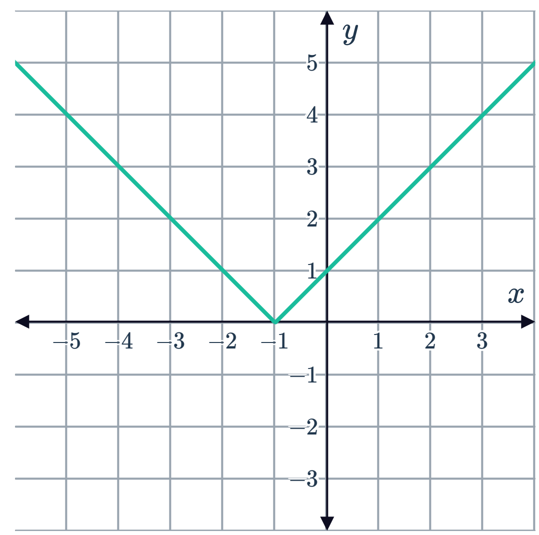
iv 2

10

a

x	-2	-1	0	1	2
y	1	0	1	2	3

b



c $x = -1$

d $(-1, 0)$

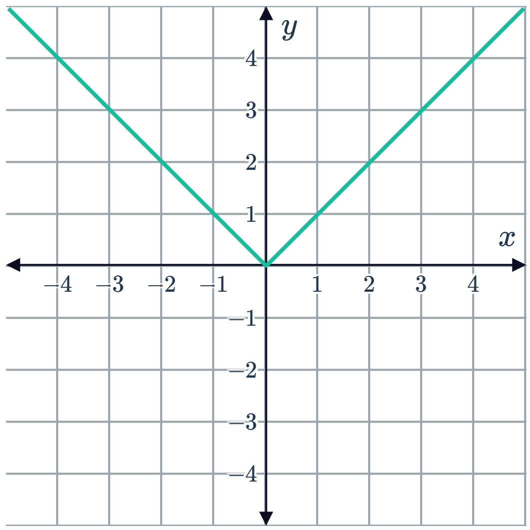
e

	Equation	Slope
$x < -1$	$y = -x - 1$	-1
$x > -1$	$y = x + 1$	1

11

a

i



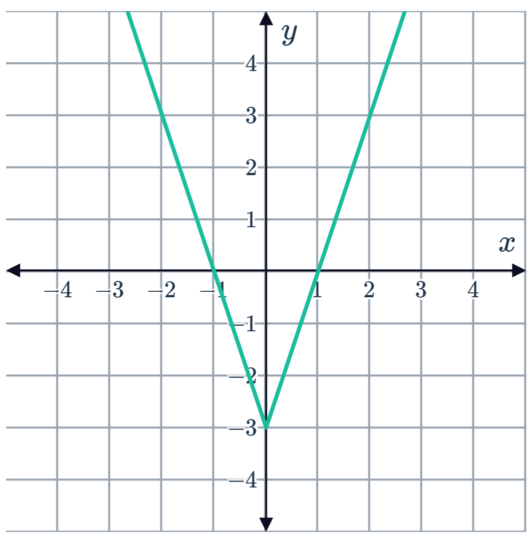
ii $(0, 0)$

iii $-\infty < x < \infty$

iv $y \geq 0$

b

i



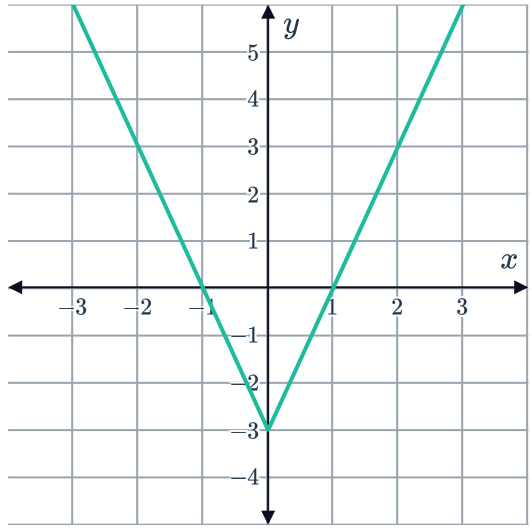
ii $(0, -3)$

iii $-\infty < x < \infty$

iv $y \geq -3$

12

a



b $x = 0$

c $(0, -3)$

d $x = -1, 1$

e $y = -3$

f -3

g 3

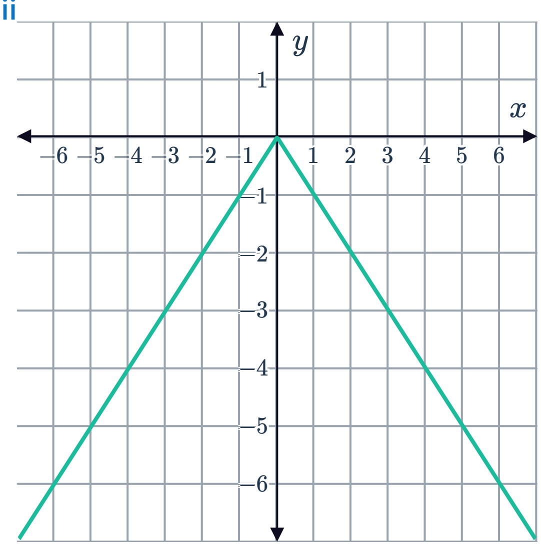
13

a

i Downwards

ii $(0, 0)$

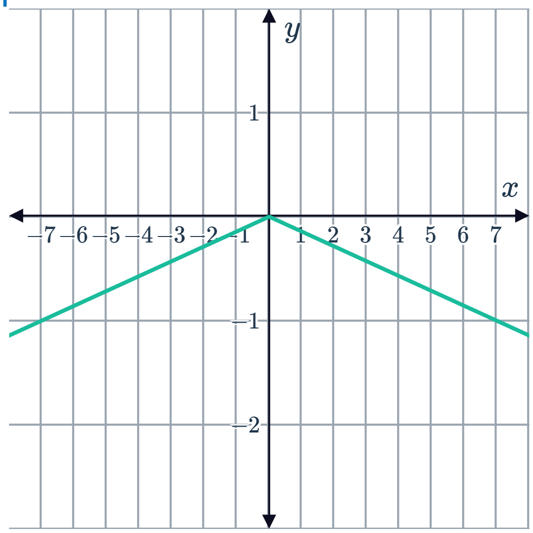
iii



b

i Downwards **ii** $(0,0)$

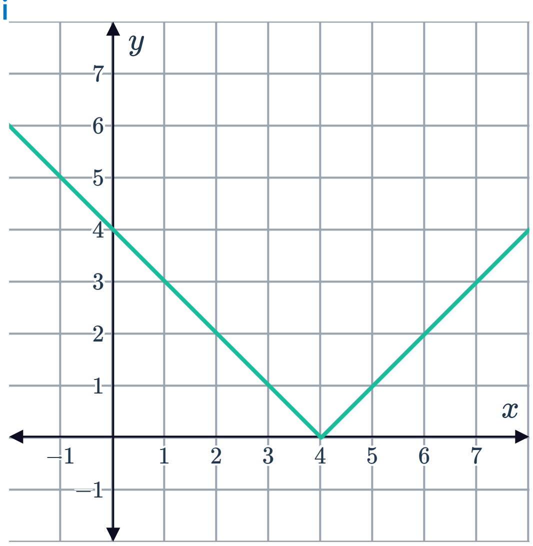
iii



c

i Upwards **ii** $(4,0)$

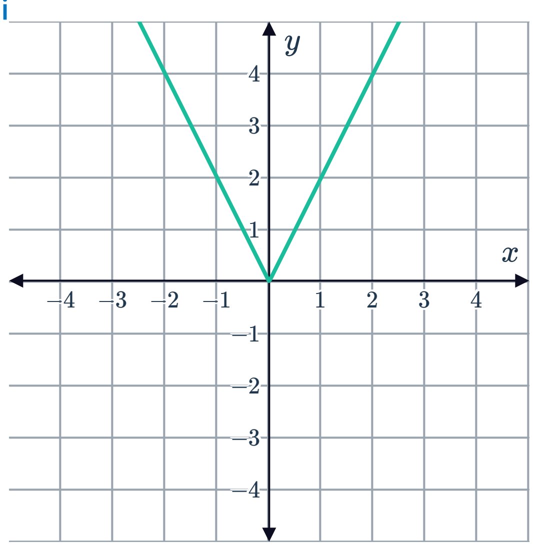
iii



d

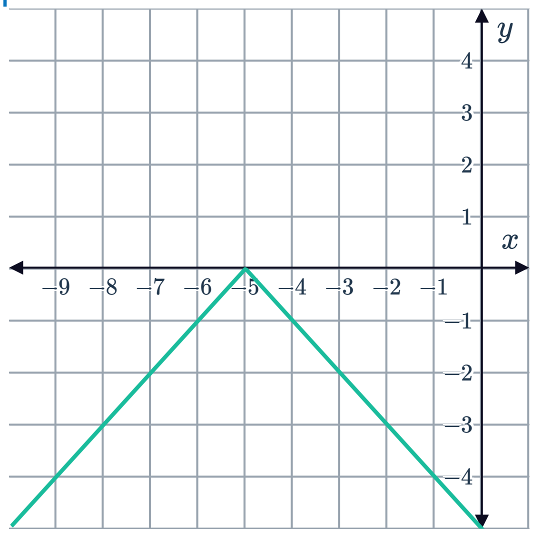
i Upwards **ii** $(0,0)$

iii



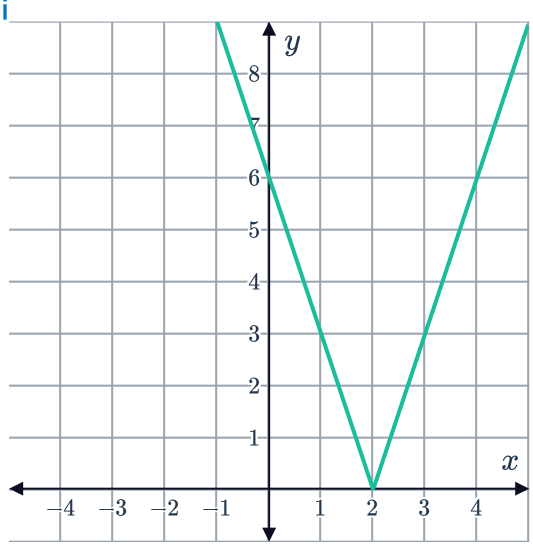
e

i Downwards **ii** $(-5, 0)$ **iii**



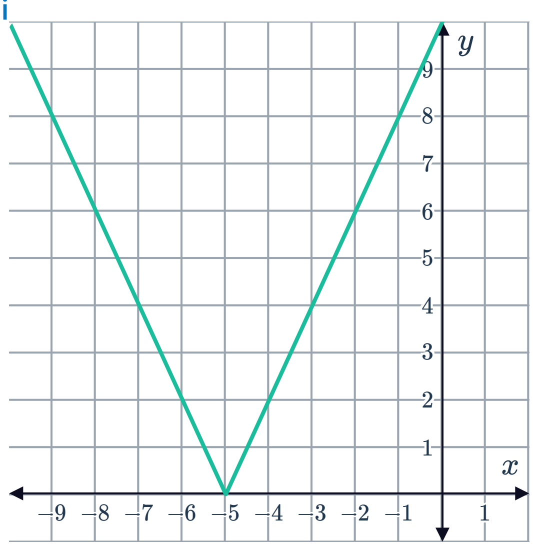
f

i Upwards **ii** $(2, 0)$ **iii**



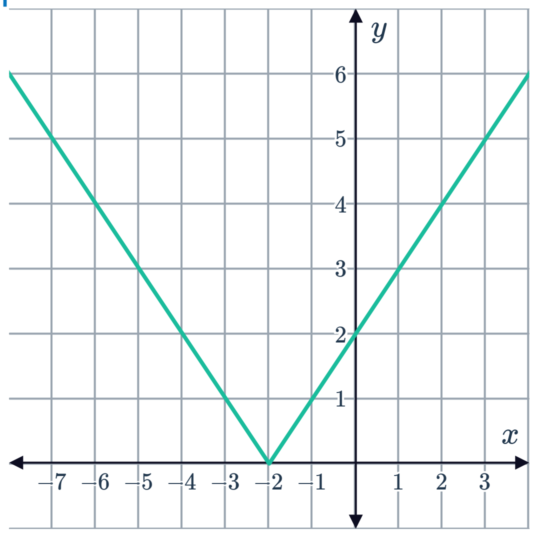
g

i Upwards **ii** $(-5, 0)$ **iii**



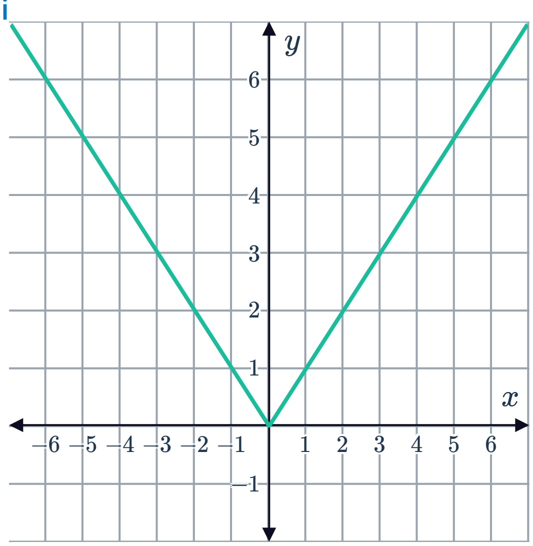
h

i Upwards **ii** $(-2,0)$ **iii**



i

i Upwards **ii** $(0,0)$ **iii**



14

a Upwards **b** $(-1,0)$ **c** $x=-1$ **d**

i No **ii** No

iii Yes

15

a Upwards **b** $(3,0)$ **c** $x=3$ **d**

i No **ii** No

iii Yes **iv** No

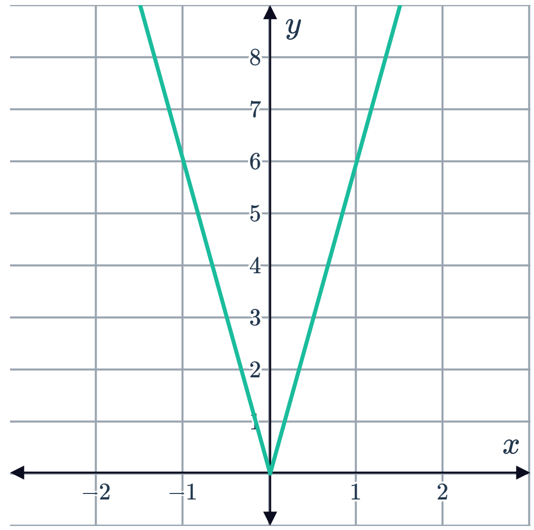
16

a Upwards

b $x = 0$

c $(0, 0)$

d



e

i No ii No iii Yes iv Yes

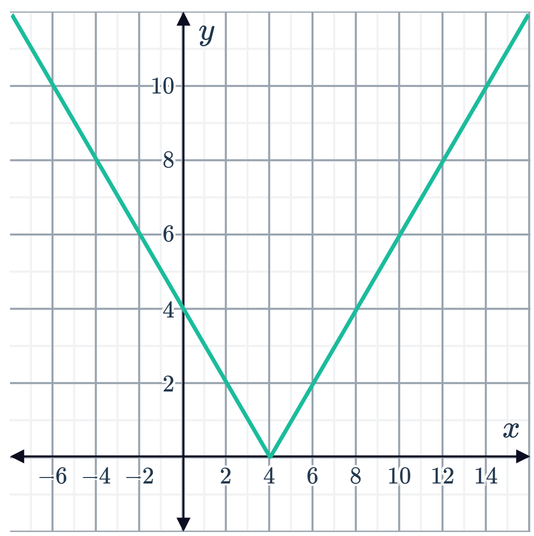
17

a Upwards

b $x = 4$

c $(4, 0)$

d



e

i Yes ii No iii No iv Yes

18

a 3

b $f\left(x\right) = \left|3\left(x - 4\right)\right|$

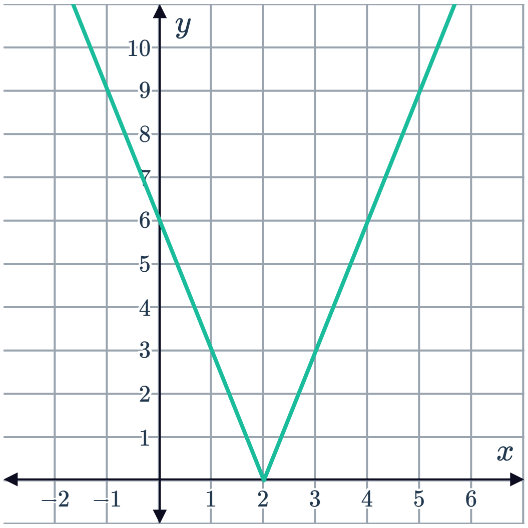
c $(-\infty, \infty)$

d $[0, \infty)$

e

19

a

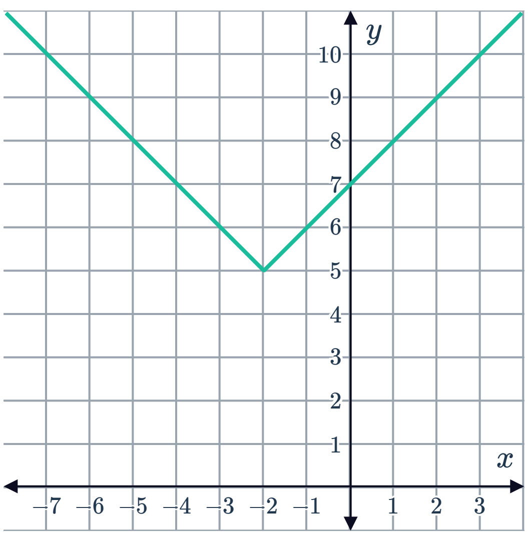


b $(2, 0)$

20

a

i

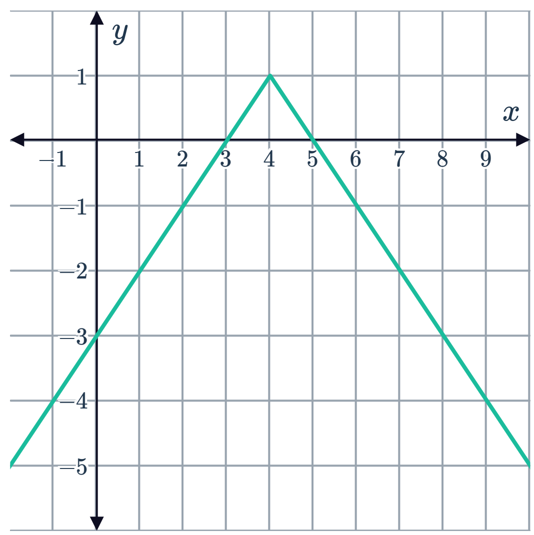


ii $(-2, 5)$

iii 5

b

i

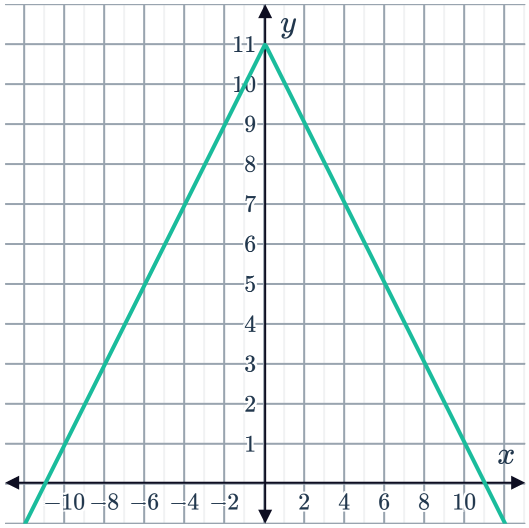


ii $(4, 1)$

iii 1

21

a



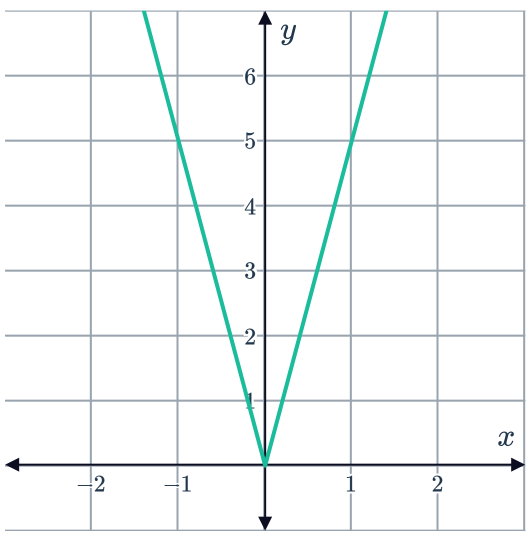
b 11

c $(-\infty, 0)$

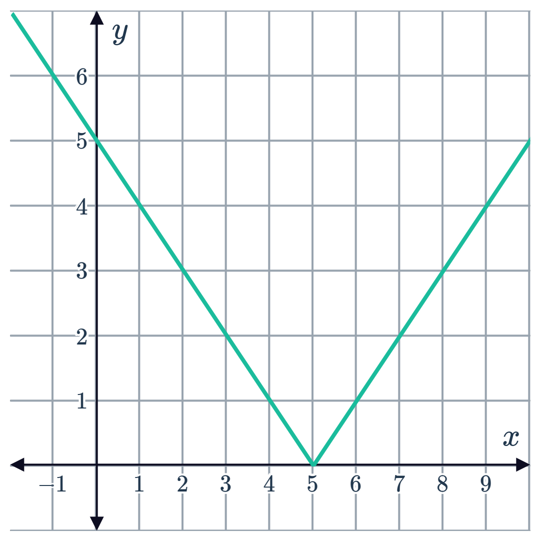
d $(0, \infty)$

22

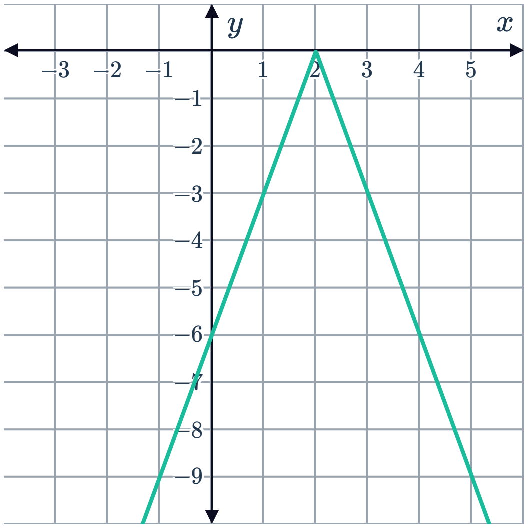
a

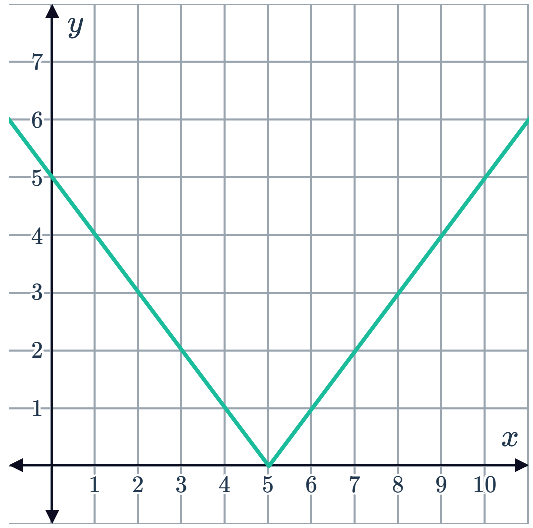


b

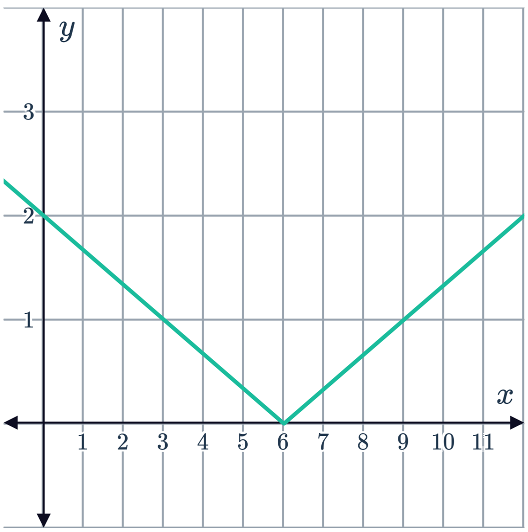


c

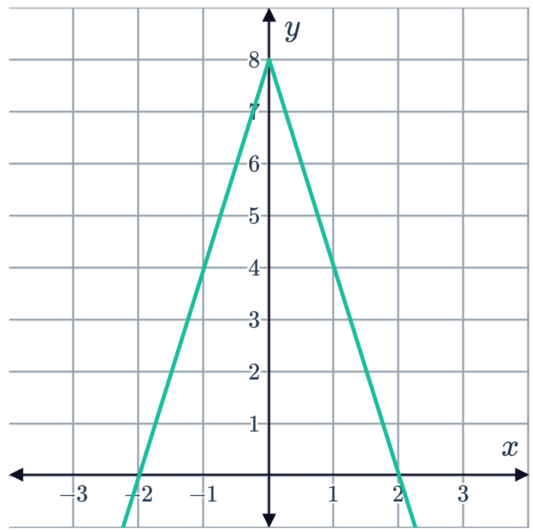

y
x
−3 −2 −1 1 2 3 4 5
−1 −2 −3 −4 −5 −6 −7 −8 −9


y
x
1 2 3 4 5 6 7 8 9 10
1 2 3 4 5 6 7

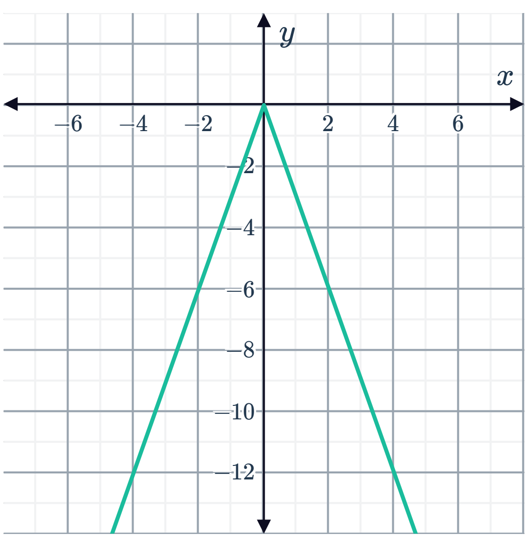
e


y
x
1 2 3 4 5 6 7 8 9 10 11
1 2 3

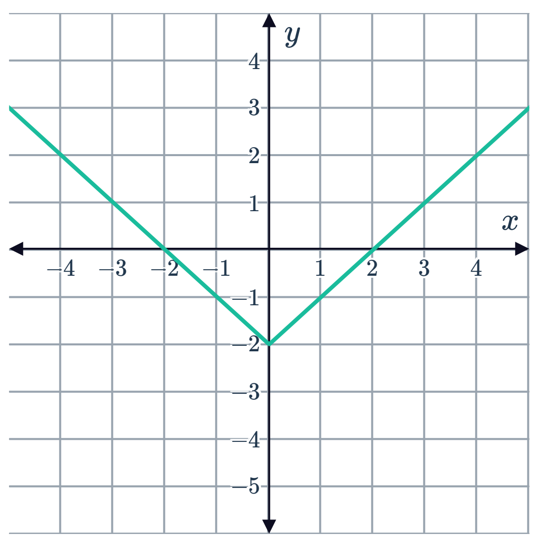
f


y
x
−3 −2 −1 1 2 3
1 2 3 4 5 6 7 8

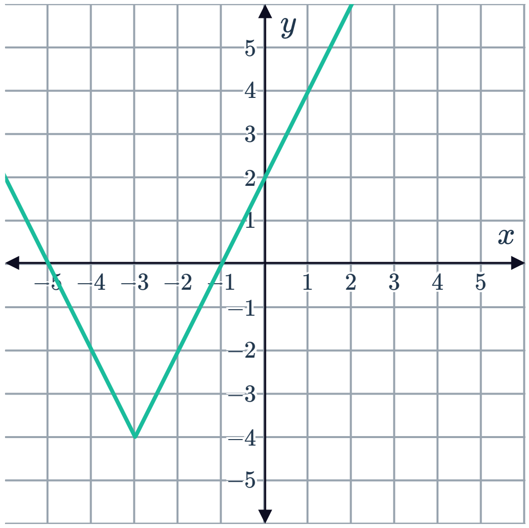
g

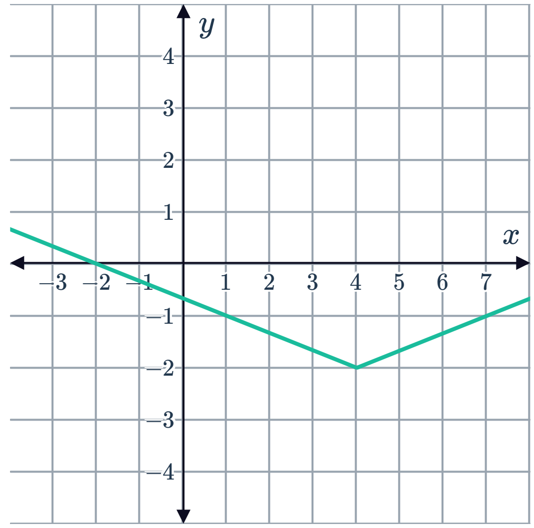

y
x
−6 −4 −2 2 4 6
−2 −4 −6 −8 −10 −12

h

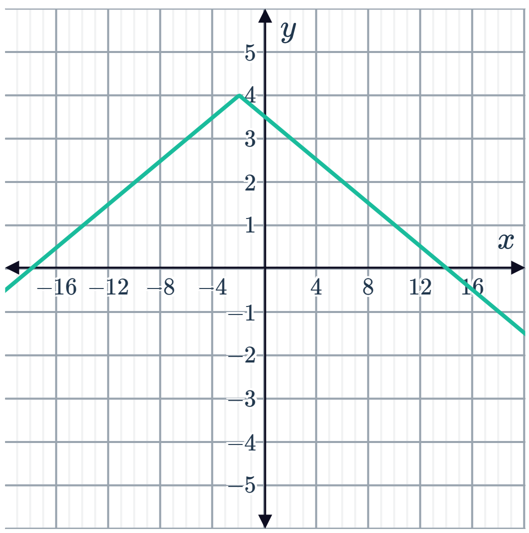

y
x
−4 −3 −2 −1 1 2 3 4
4 3 2 1 −1 −2 −3 −4 −5

i


y
x
5
4
3
2
1
−5 −4 −3 −2 −1
1 2 3 4 5
−1
−2
−3
−4
−5

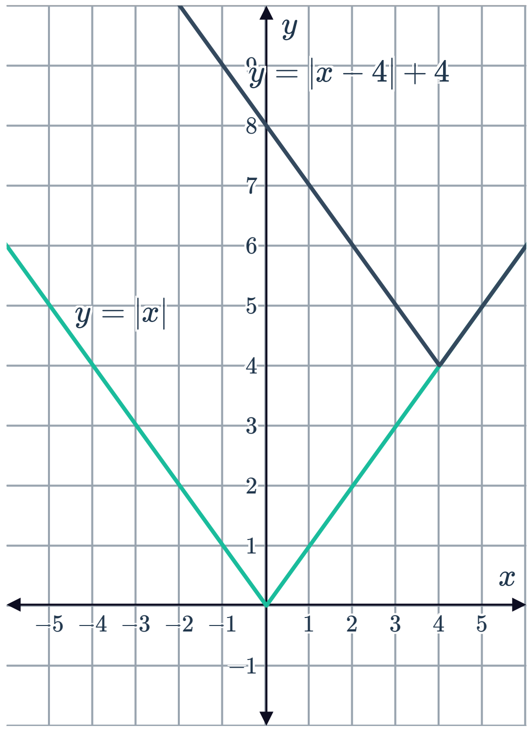

y
x
4
3
2
1
−3 −2 −1
1 2 3 4 5 6 7
−1
−2
−3
−4

k


y
x
5
4
3
2
1
−16 −12 −8 −4
4 8 12 16
−1
−2
−3
−4
−5

Transformations

23



24 $f\left(x\right) = |x+2|$

25

a $y = |x+5| - 3$

b $y = -4\left|x\right|$

c $y = -\dfrac{|x+3|}{4} + 3$

26

a Horizontal translation

b Yes

c Yes

27 Translation of 4 units down

28 Vertical stretching by a factor of 3.

29

a Reflection about the x-axis and translation of 4 units up.

b $f(x) = 4 - |x|$

30 a 3

b -3

c $(0,0)$

d $(2,-3)$

e Vertical stretching by a factor of 3, translation of 2 units to the right and 3 units down

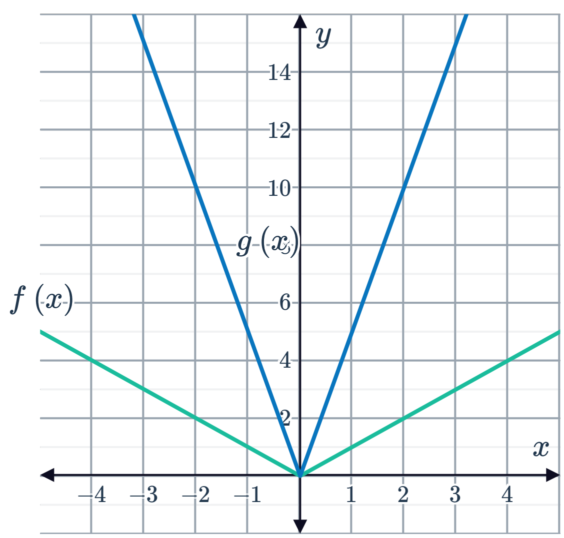
f $f(x)=3|x-2|-3$

31

a

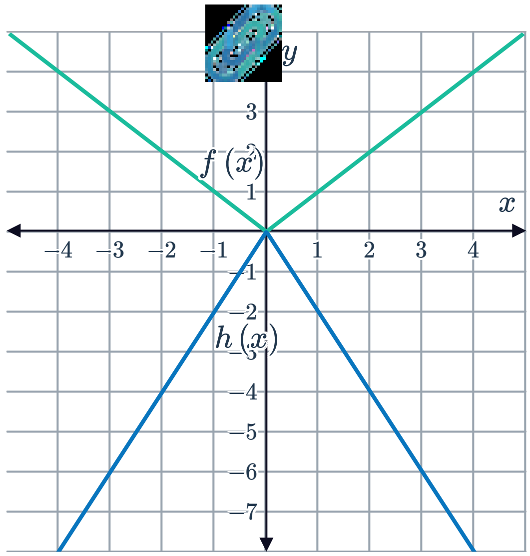
x	-3	-2	-1	0	1	2	3
$y=f(x)$	3	2	1	0	1	2	3
$g(x)=5f(x)$	15	10	5	0	5	10	15
$h(x)=-2f(x)$	-6	-4	-2	0	-2	-4	-6

b



c vertically stretched by a factor of 5

d



e Vertically stretched by a factor of 2 and reflected about the x-axis

Applications

32

a 80 mi

b 8 minutes

c Slower

33

a $0 \leq x \leq 60$

b 120 km

c 30 minutes

d Faster, because the lines are steeper than in the first graph.

34

a 700 mi/h

b $t = \left| \frac{3}{35}x - 90 \right|$

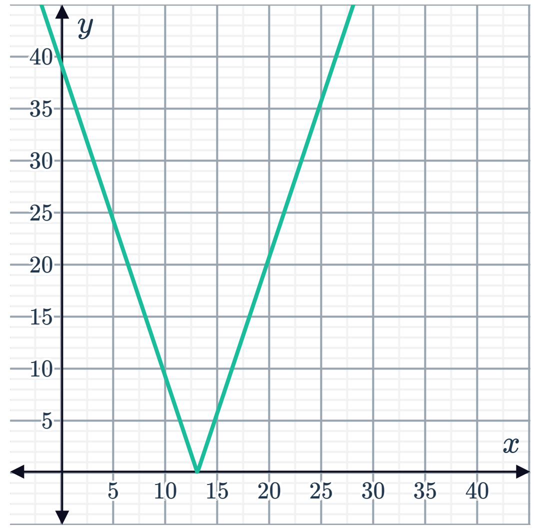
c 90 minutes

d 2.5 hours

35

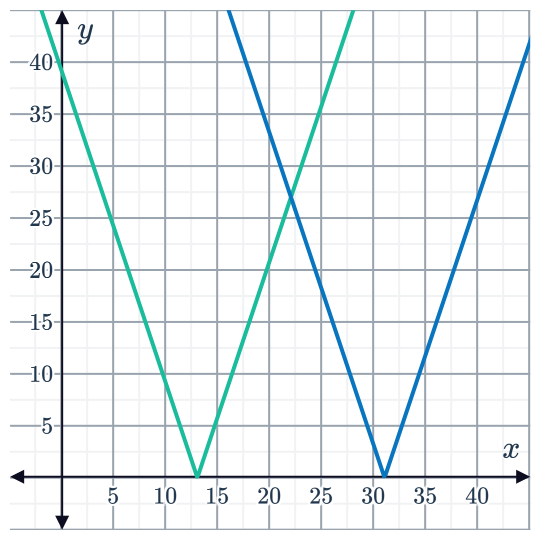
a $y = 3\left|x - 13\right|$

b



c $y = 3\left|x - 31\right|$

d



e

i Yes **ii** No **iii** Yes **iv** No